

Cherry Mittal

312-972-5734 | cherrymittal998@gmail.com | [linkedin/cherry-mittal](https://www.linkedin.com/in/cherry-mittal) | [github/cherry-mittal](https://github.com/cherry-mittal) | Chicago, IL

Professional Summary

Software Engineer with 6+ years of experience, including 4 years specializing in production-grade AI/ML architectures. Specialist in fine-tuning and deploying LLMs (BERT, Llama) for sentiment analysis and real-time summarization at enterprise scale. Expert in optimizing model latency and API performance, consistently delivering 40%+ gains in operational efficiency and system reliability.

Technical Skills

Languages: Python, SQL, PostgreSQL, Apex, Javascript, HTML, CSS

ML & AI Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NLTK, Hugging Face, Transformers, LangChain, OpenAI, ONNX

Generative AI & LLMs: GPT-4, Llama-3, BERT, Whisper, PaliGemma, RAG, LoRA / PEFT, Fine-tuning, Prompt Engineering

Tools & Technologies: Azure, Salesforce (Sales Cloud, Service Cloud, Health Cloud, Experience Cloud), REST/GraphQL APIs, FastAPI, Github, Docker, Kubernetes, Jenkins, Vector Databases(Pinecone)

Concepts: Supervised/Unsupervised Learning, CNNs, RNNs, LSTMs, NLP, Computer Vision, Feature Engineering, Model Evaluation

Experience

AI/ML Consultant (Contract)

August 2025 – Present

Billion Peoples Technologies

Remote

- **Architected an LLM-powered Text-to-SQL engine** using Python, LangChain, and PostgreSQL, automating complex data retrieval for 50+ users and **slashing manual SQL reporting overhead by 50%**.
- **Engineered an automated analytics pipeline** that converts unstructured natural language into validated SQL schemas, improving query accuracy and accelerating marketing/sales performance insights by **30%**.

Senior Software Engineer

Feb. 2022 – Apr. 2025

Optum (United Health Group)

Bangalore, India

- **CAREVOICE: Engineered a real-time call intelligence engine** leveraging Whisper STT and Llama-3, automating agent assistance and summarization to **cut post-call documentation time by 40%**.
- **Developed BERT-based sentiment analysis pipeline**, delivering sub-second agent insights and improving customer satisfaction scores (CSAT) by **20%** across high-volume service centers.
- **Developed a supervised ML classifier** (Python/Scikit-learn) to identify high-risk insurance members, enabling targeted outreach for 10K+ users and **reducing case resolution time by 55%**.
- **Re-engineered backend API architecture from REST to GraphQL**, eliminating data over-fetching and **slashing processing latency by 50%** for 1,000+ concurrent users.
- Mentored junior engineers, guiding architecture decisions, drove Agile sprint execution, and technical execution across ML and AI initiatives.

Software Engineer

Jan 2019 – Feb 2022

TEKSystems Global Services

Hyderabad, India

- **Architected Salesforce LWC applications** integrated with 30+ enterprise systems, enabling real-time bi-directional data flow for critical healthcare records.
- Directed a 6-engineer team to launch a VIP insurance platform, optimizing data processing logic to achieve a **40% reduction in system latency**.
- Implemented OIDC-based SSO, enabling secure access for 1,000+ users and improving login success rates by 20%.

Selected Projects

Mini GPT - Transformer Language Model from Scratch | *Python, PyTorch, CUDA*

Dec 2025 - Feb 2026

- Engineered a custom GPT-style autoregressive model using PyTorch, implementing multi-head self-attention and layer normalization to achieve a 1.47 cross-entropy loss on the TinyShakespeare dataset.
- Developed a high-performance training pipeline with tokenization and automated loss tracking, optimizing GPU memory utilization by 25% through vectorized operations and efficient text ingestion.

Multimodal Image-Text Classifier - PaliGemma Fine-Tuning | *PyTorch, Hugging Face, LoRA*

May 2025 - June 2025

- Fine-tuned a PaliGemma VLM using LoRA (Low-Rank Adaptation) for domain-specific image-text classification, achieving a 12% increase in F1-score while training less than 1% of total model parameters.
- Engineered an end-to-end multimodal pipeline using Hugging Face Accelerate, optimizing data loading and prompt formatting to reduce training wall-clock time by 35%.

Awards and Certifications

Optum: Spot Award at Optum - July, 2022, Diamond Award at Optum - March 2023

Certifications: Nanodegree in Generative AI - Udacity, Salesforce Certified Administrator, Salesforce Certified Platform Developer - I, Salesforce Certified App Builder

Education

Acropolis Institute of Technology

Bachelor of Engineering - Computer Science

Indore, India

July 2015 - June 2019